

FICTITIOUS PRODUCT — DEMONSTRATION DATA

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TECHNICAL SPECIFICATION DATASHEET

Compact DIN Rail Programmable Logic Controller with 4–20 mA Analog Inputs and Modbus RTU Slave

SKU:	CN-DEMO-PLC-DIN-420-MR1	Model:	CN-DIN-PLC-A1
Category:	Industrial Automation & Control / Compact PLCs	Manufacturer / Rev:	ControlNet Demo Systems Rev 1.0 (Demonstration)

Section 1: Identification and Model Overview

The industrial micro-PLC model **CN-DIN-PLC-A1** (SKU code: **CN-DEMO-PLC-DIN-420-MR1**) is a compact acquisition and logic control station for electrical distribution enclosures. It integrates embedded processing for digitizing standard analog current loops and remote supervisory telemetry over serial industrial buses.

Identification Parameter	Catalog Specification
SKU Code	CN-DEMO-PLC-DIN-420-MR1
Model Designation	CN-DIN-PLC-A1
Product Family	DIN-Logic Series Compact (Demonstration Environment)
Control Architecture	Decentralized logic acquisition unit with real-time clock and non-volatile EEPROM memory
Application Scope	Thermal process monitoring, fluid pumping automation, and serial bus telemetry

Section 2: Physical Mounting and Mechanical Form Factor

Modular industrial device engineered for direct snap-on mounting on standard interior distribution rails.

Mechanical Parameter	Nominal Specification and Mounting Attributes
Mounting Type	35 mm symmetrical DIN rail mounting in accordance with IEC / EN 60715 (compatible with TH35-7.5 and TH35-15 profiles)
Retention Mechanism	Rear spring-loaded locking catch with release latch operated via flat-head screwdriver
Overall Dimensions	90 mm (Height) × 70 mm (Width, exact 4 standard DIN modules) × 58 mm (Depth)
Net Weight / Protection	210 g / IP20 ingress protection rating per IEC 60529 (for protected control panel interiors only)
Thermal Clearance	25 mm clear ventilation spacing above and below the unit relative to cable ducts or adjacent modules

Section 3: Electrical Power Supply & Operational Tolerances

Direct current power circuitry optimized for industrial switched-mode power supplies with galvanic isolation.

Electrical Parameter	Operational Requirement and Tolerance Range
Nominal Supply Voltage	24 VDC (regulated direct current)
Operating Voltage Range	18.0 VDC to 30.0 VDC continuous (ripple tolerance Vpp < 5%)
Power Consumption	4.5 W maximum (with active serial communications and polarized input loops)
Power Terminal Block	Pluggable screw terminal connector, permissible conductor cross-section: 0.5 to 2.5 mm² (AWG 20–14)
Circuit Protection	Series diode reverse-polarity protection, self-resetting PTC thermal fuse, and 1500 VAC dielectric isolation

Section 4: Analog & Discrete I/O Interfaces and Sensors

Field signal configuration. Equipped with two standard analog current loops and deliberately lacking modulating analog outputs.

Channel Type	Provision	Technical Specification & Signal Characteristics
Analog Inputs	2 channels (AI1, AI2)	Passive 4–20 mA current loop; internal 250 Ω precision shunt; 12-bit ADC resolution (4096 counts); overall accuracy ±0.2% full scale; configurable digital software filtering
Analog Outputs	0 channels (NONE)	NO ANALOG OUTPUTS AVAILABLE. The unit lacks DAC circuitry and provides no 4–20 mA or 0–10 V output drivers
Discrete Inputs	4 channels (DI1–DI4)	24 VDC optically isolated discrete inputs (sink / PNP type), 5 mA nominal burden per channel at 24 V
Discrete Outputs	4 channels (DO1–DO4)	Electromechanical dry relay contacts SPST-NO (250 VAC / 30 VDC, 2 A maximum resistive load)

Section 5: Communications, Fieldbus and Protocol Specifications

Industrial serial communication port for supervisory telemetry and hierarchical automation network integration.

Network Parameter	Supported Configuration and Protocol Details
Physical Bus Interface	1 × RS-485 half-duplex (2-wire: terminals A/D+, B/D-, and isolated GND), with jumper-selectable 120 Ω bus termination
Industrial Protocol	Modbus RTU in SLAVE (Server) mode. Configurable slave node address: 1 to 247
Baud Rate Options	Selectable via software registers or DIP switches: 9600, 19200, 38400, 57600, and 115200 bps (default: 19200 bps)
Serial Frame Format	8 data bits, selectable parity (Even, Odd, None), 1 or 2 stop bits (factory default: 8-E-1)
Supported Modbus Functions	Supported function codes: 01 (Read Coils), 02 (Read Discrete Inputs), 03 (Read Holding Regs), 04 (Read Input Regs: AI1/AI2)
Modbus TCP / Ethernet	NOT SUPPORTED. The device lacks an Ethernet controller, RJ-45 physical port, and onboard TCP/IP stack

Section 6: Engineering Constraints and Design Contraindications

Mandatory technical exclusion criteria and design constraints for system engineering and integration:

Critical Constraint / Contraindication	Technical Rationale and Design Exclusion Rule
CRITICAL CONTRAINDICATION 1: DO NOT CLAIM OR DESIGN MODBUS TCP	The CN-DIN-PLC-A1 does NOT possess an Ethernet interface and does NOT support Modbus TCP . Under no circumstances should direct IP network connectivity to Ethernet SCADA networks be assumed. Any integration into TCP/IP packet networks strictly requires an external RS-485 to Modbus TCP gateway or converter.
CRITICAL CONTRAINDICATION 2: NO ANALOG OUTPUT AVAILABLE	This equipment possesses NO analog control output (neither 4–20 mA nor 0–10 V) . It is strictly contraindicated to specify this SKU for direct modulating control of variable frequency drives (VFDs), electropneumatic valve positioners, or continuous actuators without adding external expansion modules.
CRITICAL CONTRAINDICATION 3: SLAVE-ONLY OPERATION	The onboard RS-485 transceiver operates strictly as a Modbus RTU slave . It has no capability to initiate master polling queries or act as a network client to query other peripheral devices.
POWER SUPPLY RESTRICTION: PASSIVE 4–20 mA INPUTS	Terminals AI1 and AI2 are passive current receivers (internal 250 Ω shunt) and do NOT inject excitation voltage. Connected external 2-wire transmitters must be powered from an external loop power supply (24 VDC).